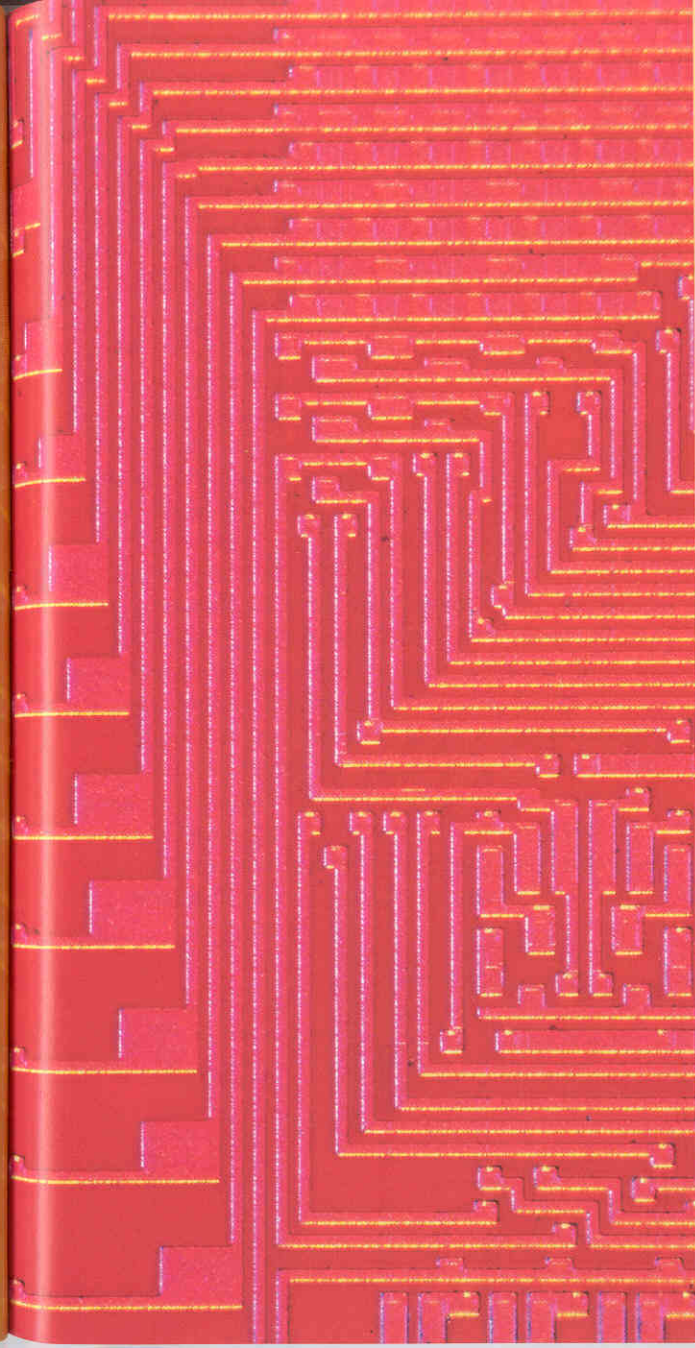


the quantum computer

A prediction was made in 1965 by Gordon Moore, the cofounder of Intel, who said that the number of transistors in computer chips would double every 18 months. This prediction, now known as “Moore’s Law,” has so far proved to be correct, but the law will soon expire, as a transistor is now being developed with components only three atoms thick. Before long, the physical limitations of using atoms to make working parts will be reached. The next logical step is to use atoms and molecules themselves to perform memory and processing tasks inside a quantum computer. Whereas computers now manipulate “bits” that exist in one of two states—0 or 1—quantum computers encode information as quantum bits, or qubits. A qubit can be a 1 or a 0, or in a superposition that is simultaneously both 1 and 0, or somewhere in between. Because a quantum computer can contain these multiple states simultaneously, it has the potential to be millions of times more powerful than any of today’s supercomputers.



soon redundant?
Part of the circuitry of a microchip, magnified 200 times. Quantum computers may soon make these seem as quaint as glowing electronic valves are today.